

C² K4 DOUBLE-MASK-OTP HYPOTHESIS

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ABSTRACT

A hypothesized on approaching the decryption of K4 from Jim Sanborn's Kryptos (1990) sculpture located within the grounds of the CIA Headquarters. Existing information on Kryptos was combined with historical research on cryptography methods, the artist Jim Sanborn, and former CIA communications officer Ed Scheidt to develop an inference model of K4 and hypothesize an approach. Based on our findings, it may be very likely K4 has been encrypted through a multi-step process involving one-time pad and several masks.

INTRODUCTION

On July 12th, 2019, we stumbled upon a YouTube video published by Great Big Story titled *Cracking the Uncrackable Code* and a thumbnail of repeating KRYPTOS characters. Immediately this caught our eyes and due to our unique skills and interests in computers/mathematics, and espionage history/art, we immediately decided to tackle his unsolved puzzle.

In our initial research of the puzzle, we learned that Jim Sanborn's Kryptos is a set of sculptures commissioned by the CIA Fine Arts Commission to coincide with the opening of the New Headquarters Building (NHB) in 1991. Comprised of multiple rock formations and sculptures, its centerpiece is a curved 4 panel copperplate screen with 1,735 characters cut out. On the right side is an encoded text and the left a series of characters arranged in the style of a 16th century Vigenere Tableau. In the encoded text, there are presumably 4 sections that have been designated K1 through K4 and Morse code messages scattered within the rock sculpture formations collectively known as K0. As

of July 15, 2019, K0, K1, K2, and K3 have been deciphered but K4 remains an unsolved mystery. K4 is a 97 or 98 set of alphabet letters that has an extremely difficult cryptographic code.

That weekend, we buckled down to embark on an internet research frenzy. We scoured through the internet to recap on decades of knowledge poured into Kryptos¹. After sifting through potential cryptographic analysis and wild theories, we decided to create a standardized logical approach to K4 using K0 through K3 as clues and identify Mr. Sanborn's original approach to encoding Kryptos. K0 to K3 were examined through a set of conditions (see below in materials and methods) and we attempted to integrate these elements to propose a multi-step decryption of K4 supported by recorded comments from Mr. Sanborn and Mr. Scheidt.

MATERIALS AND METHODS

Without the advent of broadband internet²³, the research would not be possible. Our weekend comprised of learning all the K0 to K3 ciphers and building a set of rules to assess the relevancy of Kryptos artistic elements.

Creating an inference model of K4 from K0 to K3

Upon reading of the official CIA story of Kryptos⁴, a Statement of Principles had 4 key thoughts. Among the key thoughts that stood out was:

"It should be worldly yet have identifiable American roots in concept, materials, representation, and so forth."

We postulated that if Kryptos was able to meet the CIA Statement of Principles and Key Thoughts in its entirety, then its individual components must also meet them. That is, K0, K1, K2, K3, and K4 must also meet the Statement of Principles and Key Thoughts to exist on Kryptos. Furthermore, if K0-K3 could be modelled to extrapolate and hypothesize a plausible solution to approaching K4.

¹ Many thanks to Elonka Dunin for her website links

² <http://www.thekryptosproject.com/>

³ <https://elonka.com/kryptos/>

⁴ <https://www.cia.gov/about-cia/headquarters-tour/kryptos/?tab=list-1>

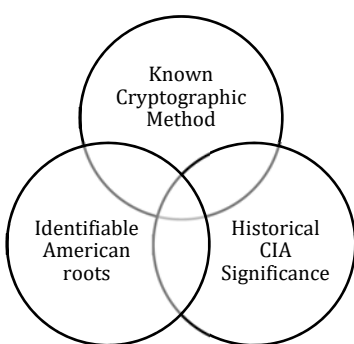
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Our model was based on meeting all three conditions:

1. A known principally manual cryptography method is employed AND
2. In the context of the CIA (and US Military), the cryptography method and/or plaintext reference was historically significant (*i.e. occurred during some era of conflict*) AND
3. Have identifiable American roots (*i.e. an American created, developed, or utilized the cryptographic method to help achieve an outcome*)

If the three conditions are met, the manual cryptography method would be used to determine the potential historical era of cryptographic importance. We will refer to meeting these three conditions as “ART” short for *American Relevance to crypTography*.

From there we would also analyze the deciphered plaintext



found to K0-K3 and identify potential patterns and hints that would lead to a logical progression of K0 to K4.

The goal is to eliminate as many possible non-relevant cryptography methods (ciphers) for K4

by using the “ART” conditions above.

Historical Era

The importance of a historical era in cryptography is that the method of encryption used should represent the appropriateness with respect to time. An example is that WPA2 WiFi encryption would not be used in the 1800's, but invisible ink would be appropriate for the 1800's.

Plaintext Hints and Original Intended Approach

On the assumption that Kryptos is a progressive puzzle from K0 to at least K4, it is very likely that hints were provided in the decrypted plaintext to provide possible guidance for future encryption sections. While K0 through K3 have already been decoded, it is possible that there will be evidence left by the artist, Mr. Sanborn, to approach K4. Evidence will likely be implied in relation to the historical

era and intentional mistakes as hints. The hint could refer to the subsequent cipher(s), or provide a fragment of the key to the cipher(s). With the hints in mind, it should lead to the original intended approach to solving KRYPTOS.

One-time Pad

The mechanics of one-time pad is an evolutionary step in substitution ciphers that can shift each individual character to any of the possible 26 A-Z letters. What differs from the prior substitution methods is the special use of keys. Firstly, the single use would make brute-force decryption of multiple messages difficult. Secondly while the key can be of any length, it would be most strongly encrypted if the key is at least the length of the plaintext and of random (unpredictable) text.

In order to utilize one-time pad, there are separate encryption and decryption methods. First, each English alphabet letter is assigned a number. A is assigned 0, B is assigned 1, C is assigned 3 all the way to Z being assigned 25.

For the encryption step, the process would be:

1. Plaintext character converted to number
2. ADD number of shifts from key (can be letter equivalent or number)
3. Divide by mod 26
4. Convert new character into equivalent letter (ciphertext)

The mod 26 will account for the shift when the letter goes past Z and will return to A, B, C, etc.

The decryption method would be the opposite:

1. Convert ciphertext into number
2. SUBTRACT number of shifts from key (can be letter equivalent or number)
3. Divide by mod 26
4. Revert back to plaintext character

Review of Supplementary Information - Dinner Notes & Interviews

The supplementary information analyzed was too small to discuss in great length, but provided additional information that may be relevant to addressing the K4 problem. We

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make the assumption that both Mr. Sanborn and Mr. Scheidt are truthful people and always make truthful statements. When ambiguous statements are made, it is assumed that the statements are still true in the original context.

K4 Ciphertext & K4 Clues

?OBKR UOXOGHULBSOLIFBBWFLRVQQPRNGKSSO TWTQSQSSEKZZWATJKLUDIAWINFBNYP VTTMZFPKWGDKZXTJCDIGKUHUAUEKCAR

For reference, it is important to also have the K4 ciphertext available. In 2010 the clue was the 64th to 69th characters NYPVTT decode to BERLIN and in 2014 the word after BERLIN (decoded) is CLOCK (decoded).

RESULTS

K0

Manual cryptography method	International Morse per memo from Jim Sanborn
CIA or USM significance	Communication method used by Lincoln, Ulysses S. Grant and Robert E. Lee to send encrypted messages during Civil War Radio telegraphy for World War I & II
Identifiable American roots	Samuel Finley Breese Morse was American and the co-developer of American Morse code (1836) for commercial use - the precursor to International Morse
Meets ART criteria?	All three conditions are met.
Plaintext	eeVIRTUALLYe eeeeeeINVISIBLE DIGETALeee INTERPRETATI-? eeSHADOWee FORCESeeeee LUCIDeee MEMORYe TISYOUR POSITION (e)? SOS RQ
Noted misspelling(s)	Digital Interpretat
Possible era reference(s)	Civil War, World War I, World War II

K1

Manual cryptography method	Quagmire III (of IV) ⁵ – a variation of Vigenere Keys – KRYPTOS (cipher key), PALIMPSEST (passphrase)
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CIA or USM significance	Plaintext (below) of subtle shading and absence of light referring to “invisible ink” or the “sympathetic stain” during the Revolutionary War of the 18 th century (stenography)
Identifiable American roots	Quagmire ciphers (four variants) are part of the Cipher Exchange in the American Cryptogram Association
Meets ART criteria?	All three conditions are met.
Plaintext	Between subtle shading and the absence of light lies the nuance of iqlusion.
Noted misspelling(s)	Iqlusion (Illusion?)
Possible era reference(s)	Revolutionary War (via plaintext)

K2

Manual cryptography method	Quagmire III (of IV) ⁶ – a variation of Vigenere Keys – KRYPTOS (cipher key), ABSCISSA (passphrase)
CIA or USM significance	Vigenere used by Confederates during American Civil War that was generally transmitted by Morse code, but would lead to the demise due to error and spacings ⁷ .
Identifiable American roots	Quagmire ciphers (four variants) are part of the Cipher Exchange in the American Cryptogram Association (same as K1)
Meets ART criteria?	All three conditions are met.
Plaintext	It was totally invisible Hows that possible? They used the Earths magnetic field X The information was gathered and transmitted underground to an unknown location X Does Langley know about this? They should Its buried out there somewhere X Who knows the exact location? Only WW This was his last message X Thirty eight degrees fifty seven minutes six point five seconds north Seventy seven degrees eight minutes forty four seconds west ID by rows*
Noted misspelling(s)	Underground and underground Correction of K2 from ID BY ROWS to LAYER TWO
Possible era reference(s)	American Civil War, World War I

⁵ <http://www.cryptogram.org/resource-area/cipher-types/>

⁶ <http://www.cryptogram.org/resource-area/cipher-types/>

⁷ <http://www.eiaonline.com/history/codecrackers.htm>

K3

Manual cryptography method	Double Columnar Transposition: transposition by 42x8 matrix rotated 90 degrees CW, transposed to 14x24 matrix and rotated 90 degrees CW
CIA or USM significance	During the American Civil War, the Union utilized a columnar transposition cipher in lieu of the more sophisticated substitution ciphers. Precursor to CIA, the SSO also utilized the double transposition cipher.
Identifiable American roots	Usage of the cipher by Anson Stager, the co-founder of Western Union ⁸ telegraph company ⁹ in 1861
Meets ART criteria?	All three conditions are met.
Plaintext	SLOWLY DESPARATLY SLOWLY THE REMAINS OF PASSAGE DEBRIS THAT ENCUMBERED THE LOWER PART OF THE DOORWAY WAS REMOVED WITH TREMBLING HANDS I MADE A TINY BREACH IN THE UPPER LEFT HAND CORNER AND THEN WIDENING THE HOLE A LITTLE I INSERTED THE CANDLE AND PEERED IN THE HOT AIR ESCAPING FROM THE CHAMBER CAUSED THE FLAME TO FLICKER BUT PRESENTLY DETAILS OF THE ROOM WITHIN EMERGED FROM THE MIST X CAN YOU SEE ANYTHING Q (?)
Noted misspelling(s)	Desparatly
Possible era reference	American Civil War, World War II

⁸ Western Union money order is the choice of Jim Sanborn's verification method of K4

⁹ <http://www.eiaonline.com/history/codecrackers.htm>

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DISCUSSION & ANALYSIS

Based on the analysis from K0 to K3, there appears to be overarching themes of cryptography through significant periods of American historical conflict. The methods used appear to have a correlation to its actual usage in history.

	American Revolution	American Civil War	World War I and/or II	Cold War
K0		✓	✓	
K1	✓			
K2		✓	✓	
K3		✓	✓	
K4				?

What is missing from K0 through K3 are any themes or references pertaining to the Cold War. While not absolutely certain, there is a high possibility that K4 will be in the Cold War era and have a cipher system relevant to the Cold War.

Two ciphers that would fit both Cold War usage and a high degree of encryption would be one-time pad or VIC cipher. Assessing both the one-time pad and VIC ciphers we have the following characteristics.

	OTP	VIC
Method	Individual numeric encoding per character on an additive basis from 0-25 (A to Z)	A straddling checkerboard for letters, numbers, and unique numeric escape characters (fillers) ¹⁰
Ciphertext display	English alphabetic characters	Numeric
Origin	Frank Miller (USA) in 1882 and re-invented in 1917 by Gilbert Vernam and Joseph Mauborgne (mechanized OTP as Vernam Cipher)	Soviet spy Reino Hayahnen codename "VICTOR"

While we do acknowledge the Vernam cipher, in our opinion, it is just a digital evolution of the one-time pad and could be better represented by a one-time pad for Kryptos. Again, by applying the ART model, the VIC cipher is a manual cryptography method, has Cold war significance, but does not have identifiable in American roots. The one-

time pad can be a manual cryptography method, has Cold War relevance to CIA/USM, and has identifiable American roots.

"BERLIN" & "CLOCK" Clues

Mr. Sanborn has been extremely generous to provide several clues in 2010 and 2014 regarding K4. He indicated to the New York Times that positions 64 to 69 and positions 70 to 74 spell BERLINCLOCK when decoded from NYPVTT and MZFPK. Using a one-time pad, the key for positions 64 to 74 would be MUYKLGKORNA as a decode key. For example, the N (value 13) would need 12 shifts backwards (i.e. letter M) to reach the letter B.

While critics may discount the one-time pad method (due to the absence of a key and the ability to create any plaintext), what is crucial is the letter frequency of K4 in terms of M and Y – there is one a single M and Y on the 97 K4 cipher text. Currently the M and Y on the cipher text are on the 70th and 65th positions respectively and somehow would have to shift to get into the 64th and 66th position respectively. If this is the case, K4 would act both as the ciphertext and public key in a one-time pad decryption. To increase the level of difficulty, the K4 public key would likely have to be rearranged.

Masking techniques

In our opinion, masking techniques differ from actual cipher methods as deterrents to computational methods. Since computers can theoretically brute force all possible combinations, the masking techniques would likely remove the increasing computing power advantage by adding non-linear human approaches. Assuming a public key on K4 is used for the one-time pad cipher, there should be at least two masks done.

One mask for the public key in terms of multiple transpositions (matching the redundant technique in K1 and K2 substitutions and the double transposition of K3). The second mask would be for the plaintext to mitigate computational brute-force by (from a 2005 NPR interview

¹⁰ https://www.cia.gov/library/center-for-the-study-of-intelligence/kent-csi/vol5no4/html/v05i4a09p_0001.htm

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with Ed Scheidt in which he eluded to the number four spelt as P-H-O-A-R¹¹.) using a phonetic masking technique.

Therefor we propose a Double-Mask-OTP Hypothesis approach, whereby:

- K4 has been encrypted using one-time pad (OTP)
- K4 is a public decryption key using OTP that is masked through multiple transpositions in a non-linear way AND
- The plaintext is masked phonetically to prevent brute-force computational decryption

Edit – July 22, 2019 – As of July 21, 2019 we believe we have decoded K4 using our proposed *C² Double-Mask OTP Hypothesis*. Details will follow. Special thanks to Elonka Dunin for her Great Big Story, the Yahoo! Kryptos group, and everyone else involved in detailing the accounts of Kryptos. At this time of writing, Jew-Lee's Kryptos project appears to be offline, possibly due to increased network traffic.

CONCLUSION

Our hypothesis is a simple, yet elegant approach to decoding K4. It would allow for the maximum amount of flexibility and creativity for Jim Sanborn to create K4. Ed Scheidt may be aware of the plaintext, but would not know the exact steps that Mr. Sanborn has used due to the natural of non-linear transpositions of the hypothesized K4 public key.

In a sense, art is very much akin to cryptography. The artist has an intent (plaintext) that is encoded through some medium (ciphertext). Individuals have to decode the artwork through various materials and methods (key) to recreate the meaning of the art piece (plaintext) and potentially some sort of esoteric enlightenment. Jim Sanborn's Kryptos should truly be considered an artistic masterpiece.

¹¹ <http://kryptools.com/hints.hm>